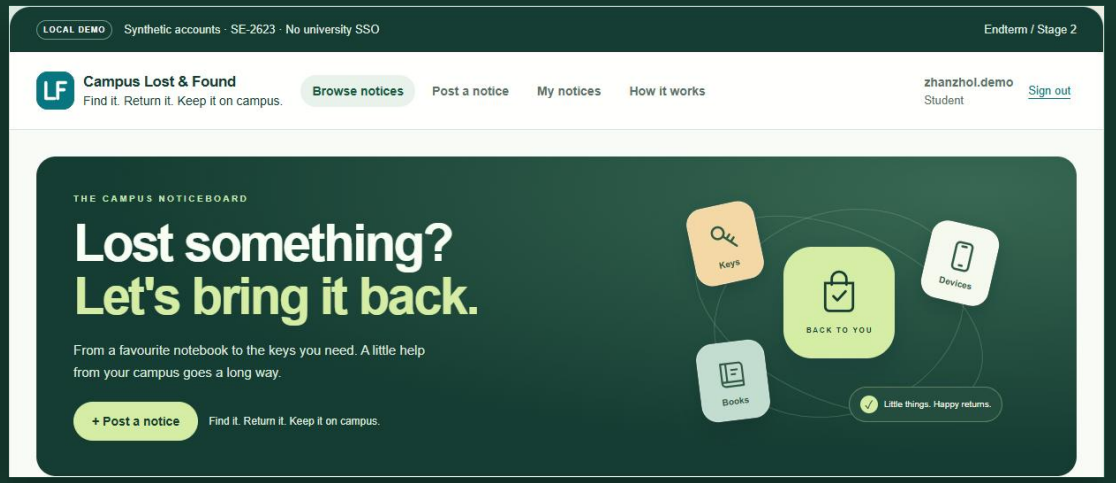
ICT FINAL PROJECT · SE-2623 · WEEK 10

Lost it on campus? Find it in one place.

Campus Lost & Found — the finished, tested system, and what three of us learned building it.



Real screenshot of the final build · synthetic demo data · runs locally

Mengdigerey Zhanzhol
Galitskiy Daniil
Bisengaliev Daulet

Lost items wait for days because notices are scattered

4 of 10

students in our survey said getting an item back took more than three days

9 of 10 said a shared, searchable board would be useful

Stage 1 survey, October 2026 · n = 10 · convenience sample of fellow students — a signal that the problem exists, not a measurement of its size.

TODAY		WITH CAMPUS LOST & FOUND
Group chats posts scroll away	→	**One board** search by title and category
Paper notes only seen in one corridor	→	**University accounts** everyone knows who posted
Guard desk no one knows it is there	→	**Email the author** one click from the notice
No status is it already returned?	→	**Returned badge** closed notices stay honest

9 of 11 Musts done; university sign-in is the gap

FUNCTIONAL (F) · F3 is the only Should

F1	F2	F3	F4	F5	F6	F7	F8
partly	done	done	done	done	done	done	done
University sign-in	Publish	One photo	Browse + details	Search + filter	Email author	Edit + returned	Admin removes

NON-FUNCTIONAL (N)

N1	N2	N3	N4
done	partly	done	done
Search ≤ 2 s	Identity + rights	360 px + keyboard	30-min demo

Why F1 and N2 are only partly done

Accounts are local. Registration only accepts an @astanait.edu.kz address, passwords are scrypt-hashed and owner/admin rights are enforced.

Microsoft sign-in code is ready (OpenID Connect + PKCE, 11 tests on simulated Microsoft), but it needs an app registration in the university's Entra tenant — only IT can create it. Until a real account signs in, it stays "partly".

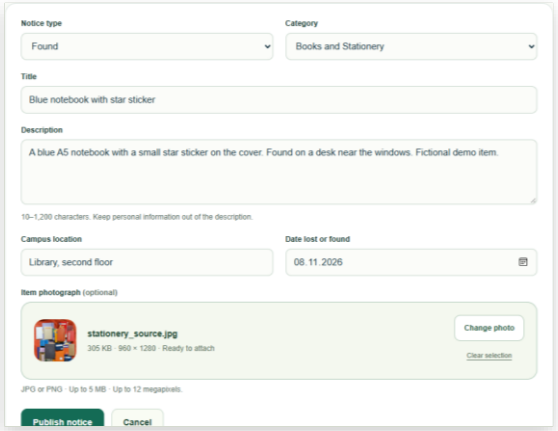
Evidence for every row

Final/Requirements/Requirements_Status.md lists the test, screenshot or measurement behind each status. All were re-run on 8 Nov 2026 on the final code.

Live demo · a student goes from “I found it” to “Returned”

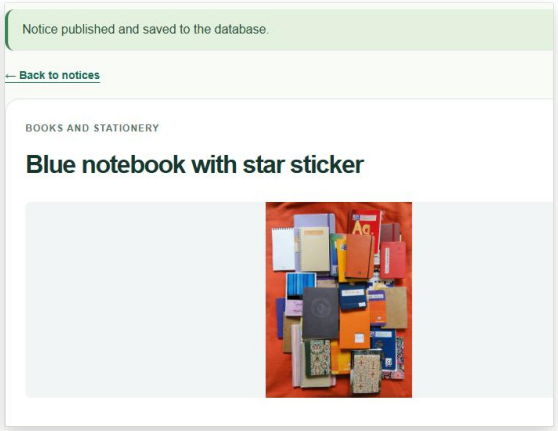
Zhanzhol · 1:15

LIVE DEMO



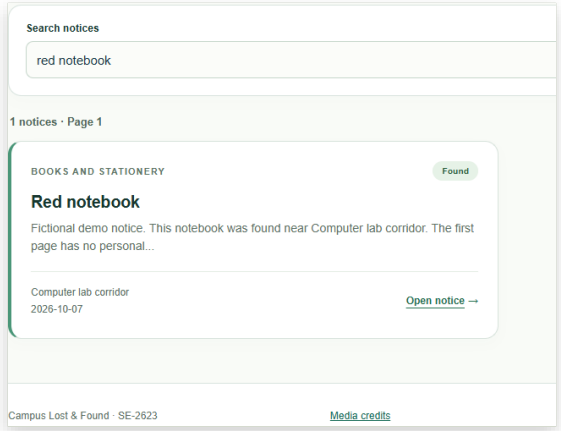
1 Publish

Type, category, title, place, date, one photo



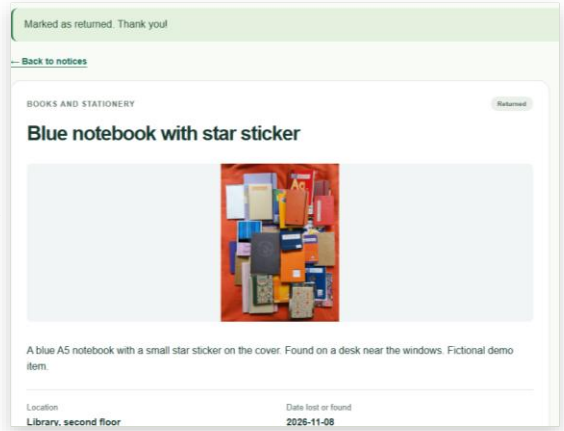
2 Saved

Stored in SQLite; photo re-encoded and private



3 Find

Search “red notebook” + category → 1 result



4 Close

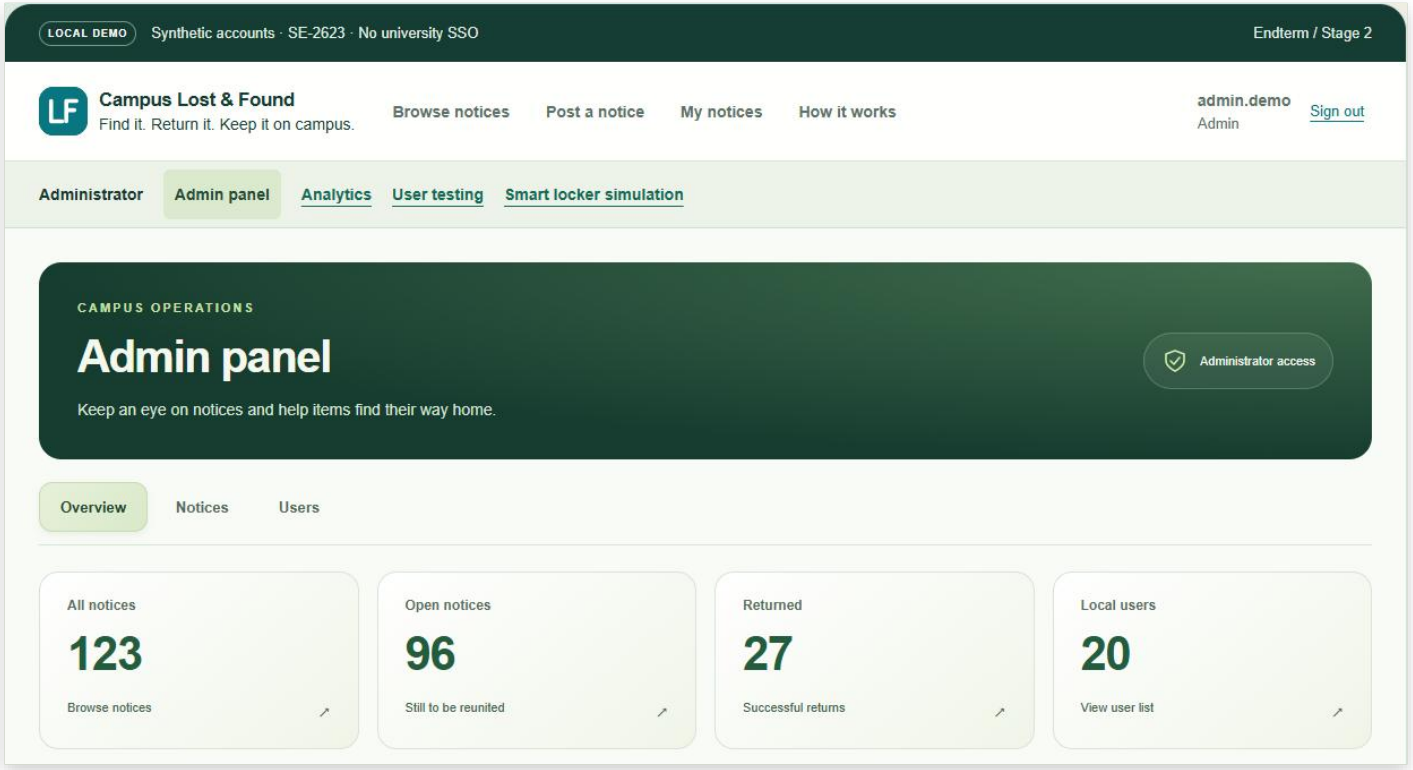
Email the author, then mark it Returned

All four frames are real captures of the final build (8 Nov 2026). If the live demo fails, talk through these frames.

Live demo · the admin keeps the board clean

Daniil · 1:10

LIVE DEMO



1 **Live counts from SQLite**
Notices, open, returned and users are counted on every visit — nothing is typed in by hand.

2 **Remove with a confirmation step**
Opening the page deletes nothing; only "Confirm removal" does. A student gets 403.

3 **Analytics on the same data**
Three SQL questions with CSV export, clearly labelled "synthetic teaching data".



Admin pages return 403 to all three student accounts and 200 to the administrator (checked 8 Nov 2026).

The final code passed every automated and security check

Daniil · 0:50

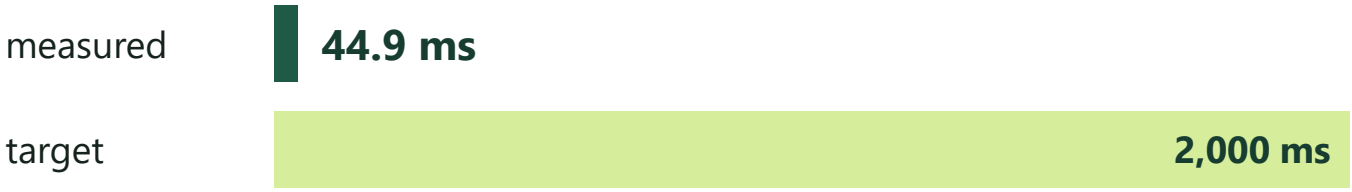
94 / 94

automated tests passed on the final code — logins, CSRF, ownership, photos, analytics, backup, locker

Ran 94 tests in 29.6 s — OK · 8 Nov 2026

These are software checks. They are never counted as human testers — that is the next slide.

N1 · Search response, 10 users at once (95th percentile)



Same scale. 100 searches from 10 signed-in sessions over local HTTP; page images and drawing time not included.

5 / 5

invalid-access attempts blocked (SC3): wrong domain, wrong password, anonymous, foreign edit, student removal

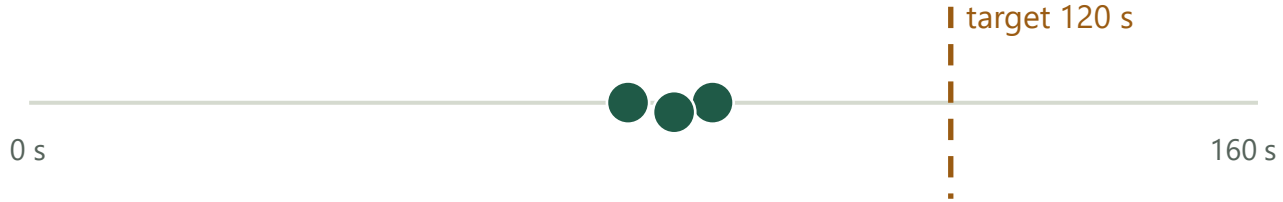
20 / 20

availability checks passed, spread over 30 minutes (N4), plus offline → retry in the browser

3 people outside the team re-tested our user criteria

Daniil · 0:45

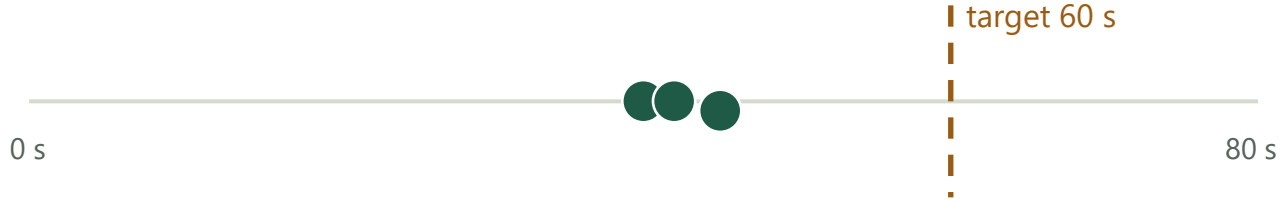
SC1 · Publish



3/3

met (≥ 80 %)

SC2 · Find



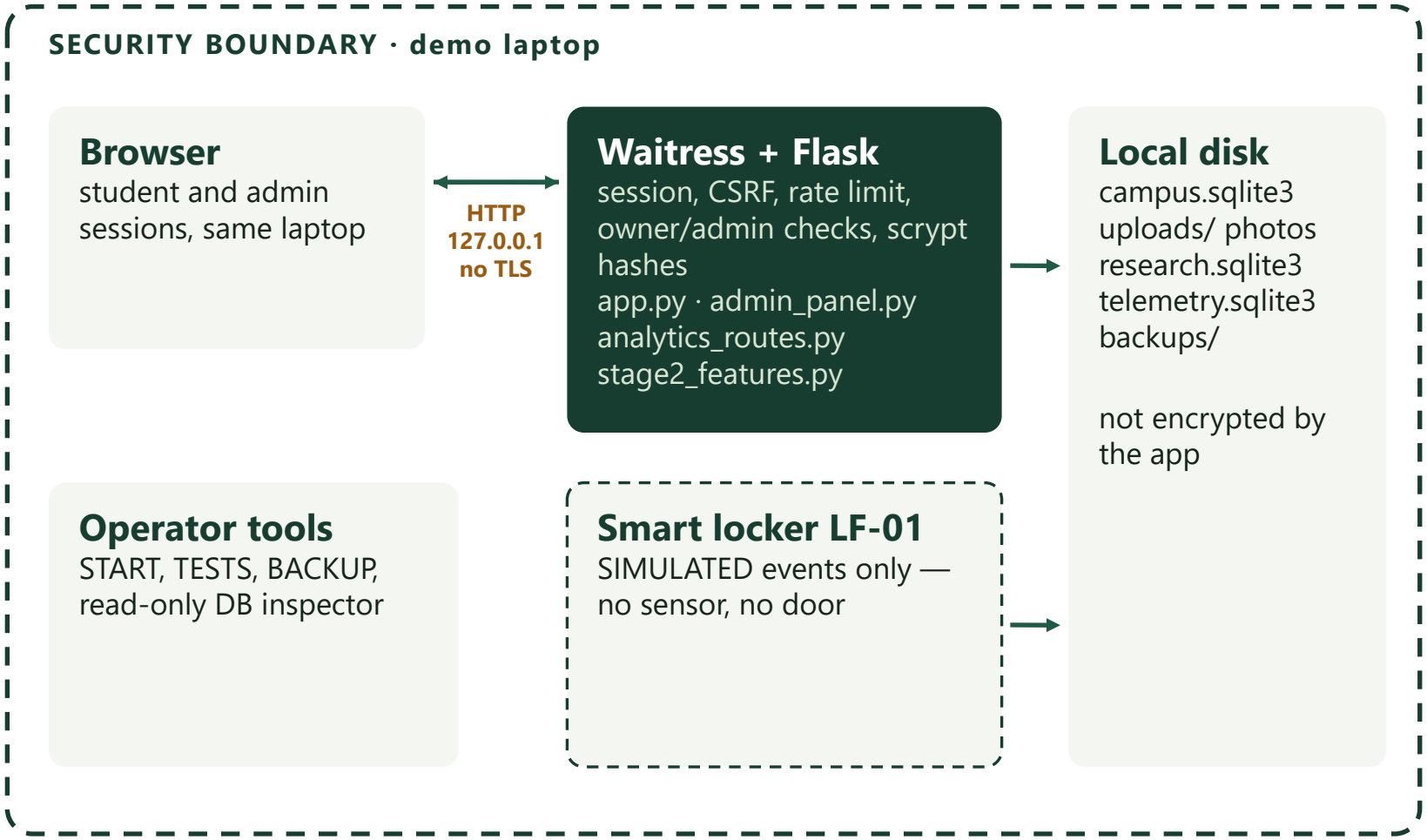
3/3

met (≥ 80 %)

Criterion	Stage 2 result	Final result	Met?
SC1 · post within 2 min (≥ 4/5)	Report: Pending · team-reported 5/5, no raw record	3/3	✓ Yes
SC2 · find within 1 min (≥ 4/5)	Report: Pending · team-reported 5/5, no raw record	3/3	✓ Yes
SC3 · invalid access blocked (5/5)	5/5	5/5 (8 Nov)	✓ Yes

Each dot is one person's first attempt (green = correct, unassisted, within target). 3 people outside the team, 2026-11-06 and 2026-11-07; Windows 11 / Chrome, Windows 11 / Microsoft Edge, Android / Chrome. Small sample: one person = 33 percentage points.

One laptop; the security boundary is its Windows account



OUTSIDE THE BOUNDARY

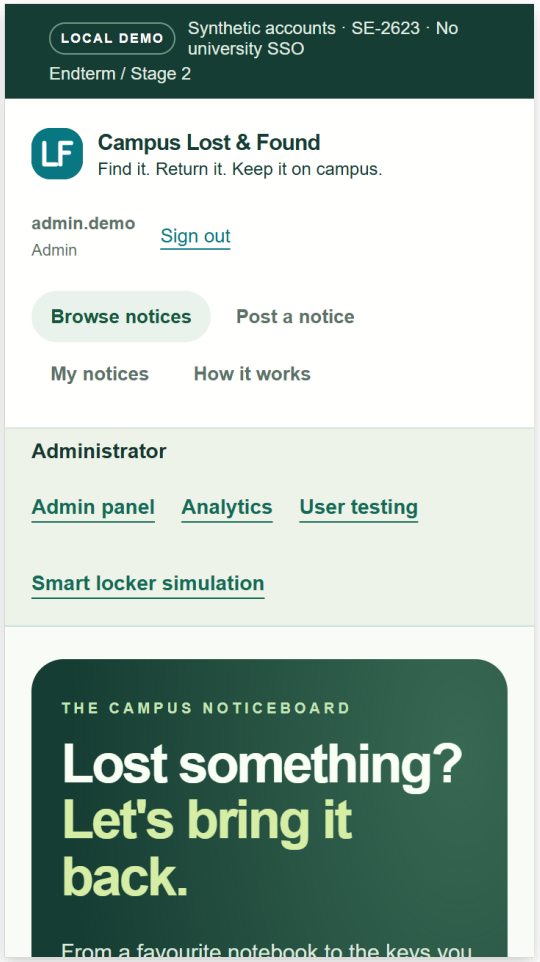
Other devices
cannot connect — server binds 127.0.0.1

Email app
"Contact author" opens a draft; site sends nothing

Microsoft Entra ID
sign-in code ready, off until IT registers the app

GitHub
public source with synthetic data only

Works on a 360 px phone, by keyboard and after a drop

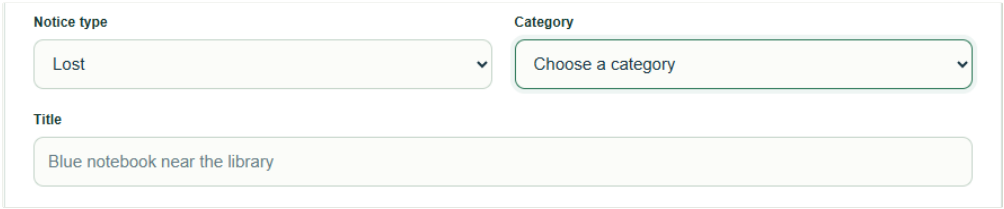


12 / 12 pages

no sideways scrolling at 360 × 760 px — student, admin, sign-in and locker pages

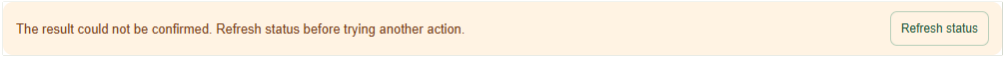
Keyboard

Tab follows the screen order to “Publish notice”; every field has a label; focus is visible



Offline → retry

network cut: a clear message, no double submit; reconnect: “Status refreshed”

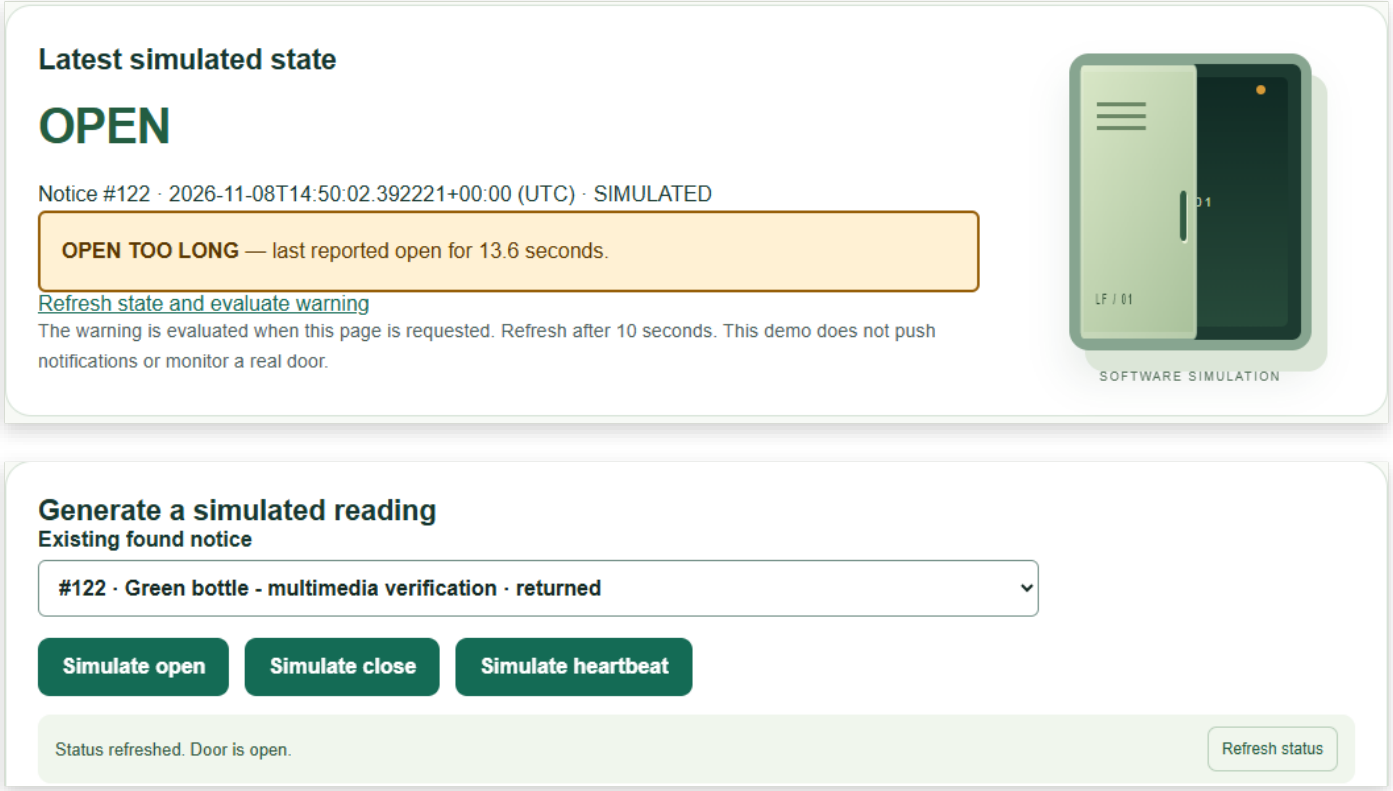


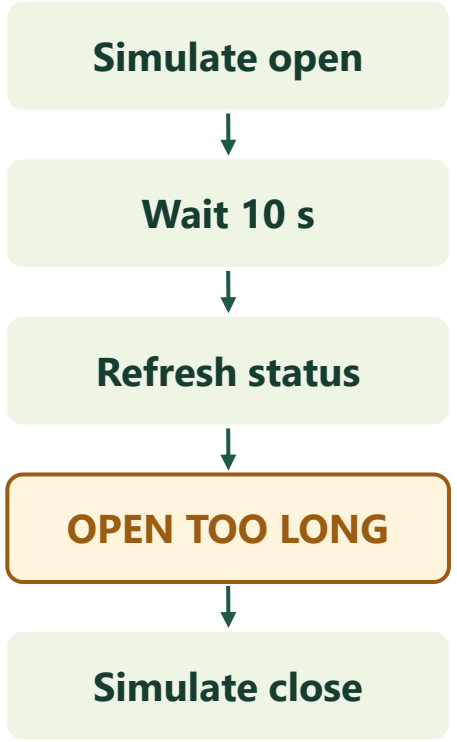
Automated checks in Microsoft Edge on 8 Nov 2026 — not an accessibility audit with real assistive-technology users.

Live demo · a simulated locker warns when it is left open

Daulet · 0:55

LIVE DEMO





SOFTWARE SIMULATION

No sensor, no door, no network device. Every reading is stored as SIMULATED and a notice is never closed automatically — a person checks the locker.

Buttons update the page in place (fetch with CSRF token, no reload). Physical version proposed: reed switch + Wi-Fi controller, HTTPS, device keys.

What we would build next, and what we learned

LIMIT NOW	NEXT STEP
University sign-in not connected	→ Code is ready: IT registers the app, we set 3 variables and test a real sign-in
One laptop, local HTTP	→ Host in a Kazakhstan data centre with HTTPS — personal data must stay in Kazakhstan
SQLite, one writer	→ PostgreSQL + object storage before 10,000 users
Free-text locations	→ A fixed list of campus places for cleaner analytics

Lessons

- Evidence beats claims — the requirement table separated "passes a test" from "a person used it".
- Cutting chat and matching kept the core flow stable; SSO is built but honestly marked "not connected".
- One upload commit hid our individual work — next time, one commit per feature.
- Start testing with real users in week 7, not week 10.

Thank you

Questions — about any part of the system — are welcome.

Repository
github.com/zondbebox/SE-2623_Endterm · branch main · commit a7e5c74

Run it
Campus_Lost_Found\START.bat → http://127.0.0.1:5055

Demo accounts (synthetic)
student zhanzhol.demo@astanait.edu.kz · admin admin.demo@astanait.edu.kz · shared demo password in the README

Mengdigerey Zhanzhol
Galitskiy Daniil
Bisengaliev Daulet

Backup · if the live demo fails, we keep going

Server stopped	Close the window, double-click START.bat again, sign in (30 s).
Port 5055 busy	Stop the old window with Ctrl+C, or run python\python.exe app.py --port 5056.
Laptop or browser fails	Narrate slides 4, 5 and 10 — every frame is a real capture of the final build.
Need motion	Local MP4 Video/SE-2623_Campus_Lost_Found_A4.mp4 (2:14) or youtu.be/76M3hSqR-7w — older interface, student flow only; say so.

Backup · where personal data lives, and why the law cares

Where it is now

Emails, password hashes and notices: data/campus.sqlite3. Photos: data/uploads/. Backups: backups/. All on the demo laptop's disk, in Kazakhstan, not encrypted by the app. GitHub holds synthetic data only.

What the law says

Law RK No. 94-V "On Personal Data and Their Protection": Art. 12(2) storage in a database located in Kazakhstan; Arts. 7–8 consent; Art. 22 protection; Art. 25(2)(8) breach notice within one working day (Russian text, as amended 14 Sep 2026).

If we went public

Kazakhstan data centre · HTTPS · university sign-in · consent text · retention period · incident procedure. We do not claim legal compliance.

Backup · with 10,000 users, access fails first, then writes

1 · Reachability
Loopback only: nobody else can open the site

→

2 · Writes
SQLite = one writer; Waitress = 4 threads; uploads up to 6 MB queue

→

3 · Storage
Photos on one disk; in-memory rate limit resets on restart

→

4 · Identity
Local accounts cannot prove 10,000 real students

Plan: host in Kazakhstan with HTTPS → PostgreSQL → object storage for photos → several app instances behind a load balancer with a shared rate limiter → Entra sign-in → load-test before launch. Our 45 ms p95 is for 10 users on a laptop and does not predict 10,000.

Backup · sources for the facts in this talk

- Stage 1 report (9 Oct 2026) — survey n = 10, requirements F1–F8, N1–N4, SC1–SC3
- Stage 2 report (1 Nov 2026) — Stage 2 statuses; SC1/SC2 recorded as Pending
- Final/Evidence (8 Nov 2026) — test_suite_final.log, search_performance_final.json, sc3_final.json, stability_30min_final.json, browser_checks.json, account_checks.json, screenshots
- Final/Testing — Final_Test_Protocol.md, final_test_results.csv, stage2_team_reported.csv
- Law RK No. 94-V, adilet.zan.kz (checked 8 Nov 2026); Microsoft Learn — Entra app registration and OpenID Connect; SQLite — Appropriate uses; OWASP cheat sheets
- Code: github.com/zondbebox/SE-2623_Endterm, commit a7e5c74 (only commit, uploaded from the zondbebox account)